

Texas Dairy Farm Research Summary

Water Costs, Energy Costs & Family Labor Production Costs by Farm Size

Week 2 Research | Vensim PLE Framework

Table 1: Water, Energy & Family Labor Expenditures by Farm Size

Cost estimates derived from USDA ARMS 2021 data, USDA NASS Texas Dairy Review 2024, MSU Extension water use studies, and University of Wisconsin Extension energy benchmarks. Texas electricity rate of ~\$0.08/kWh applied.

Cost Category	Small Farm (<500 cows)	Mid-Size Farm (500-2,000 cows)	Large Farm (>2,000 cows)
WATER COSTS			
Water use per cow per day	~30-35 gal	~28-32 gal	~25-30 gal
Annual water use (est. total)	~5.5-6.4M gal/yr	~15-23M gal/yr	~37-54M gal/yr
Water cost (pumping/treatment)	\$0.01-\$0.03/gal (~\$55K-\$192K/yr)	\$0.01-\$0.03/gal (~\$150K-\$700K/yr)	\$0.01-\$0.03/gal (~\$370K-\$1.6M/yr)
ENERGY COSTS			
Electricity use per cow per year	800-1,200 kWh	700-1,000 kWh	600-900 kWh
Estimated annual electricity cost	~\$32K-\$96K/yr (at ~\$0.08/kWh TX rate)	~\$90K-\$400K/yr	~\$300K-\$1.4M/yr
Utility costs as % of milk production cost	<2% of total cost	<2% of total cost	<2% of total cost
Fuel, lube & electricity per cwt milk	~\$1.50-\$2.50/cwt	~\$1.00-\$1.75/cwt	~\$0.60-\$1.00/cwt
FAMILY / UNPAID LABOR COSTS			
Unpaid labor as % of total labor cost	~97% unpaid (family-run)	~40-60% unpaid	~5% unpaid (mostly hired)
Unpaid labor cost per cwt milk	~\$10-\$14/cwt	~\$3-\$6/cwt	~\$0.50-\$1.50/cwt
Total cost of production per cwt (all costs)	~\$42.71/cwt (2021 ARMS, <50 cows)	~\$24-\$28/cwt (est.)	~\$19.14/cwt (2021 ARMS, 2,000+ cows)
Hired labor cost per cwt	~\$1-\$2/cwt	~\$3-\$5/cwt	~\$5-\$7/cwt
Net returns (avg.)	Negative (most years)	Near breakeven to slightly negative	Positive (~\$1.12/cwt avg. 2005-18)

Table 2: Vensim PLE Stock & Flow Framework

Element Type	Vensim Symbol	Variable Name	Description
STOCK	Rectangle	Herd Size (cows)	Total milk cows on farm; changes with births, purchases, culls, deaths
STOCK	Rectangle	Cash on Hand (\$)	Farm liquidity; changes via revenue inflows and cost outflows
STOCK	Rectangle	Cumulative Water Used (gal)	Accumulated water consumed; driven by daily water use rate per cow
FLOW (+)	Valve/Arrow ->	Births / Heifer Additions	Rate adding to Herd Stock; depends on reproduction rate and purchases
FLOW (-)	Valve/Arrow <-	Culls & Deaths	Rate removing from Herd Stock; depends on herd health, profitability
FLOW (+)	Valve/Arrow ->	Milk Revenue	Flow into Cash; = milk price (\$/cwt) x milk production (cwt)
FLOW (-)	Valve/Arrow <-	Water Cost	Flow out of Cash; = water use rate x pumping/treatment cost per gallon
FLOW (-)	Valve/Arrow <-	Energy Cost	Flow out of Cash; = kWh used (by herd size) x electricity rate (\$/kWh)
FLOW (-)	Valve/Arrow <-	Family Labor Cost	Flow out of Cash (opportunity cost); unpaid hours x off-farm wage rate. Declines sharply as farm grows.
FLOW (-)	Valve/Arrow <-	Feed Cost	Largest operating outflow; purchased + homegrown feed per cow x herd size
AUXILIARY	Circle/Arrow	Water Use Rate (gal/cow/day)	Converter: ~25-35 gal/cow/day depending on farm size and season
AUXILIARY	Circle/Arrow	Electricity Use (kWh/cow/yr)	Converter: 600-1,200 kWh/cow/yr; lower on larger, more efficient farms

Suggested Vensim PLE Model Structure

Stock 1 — Herd Size (cows): Inflows: Births, Heifer Purchases. Outflows: Culls, Deaths.

Stock 2 — Cash on Hand (\$): Inflows: Milk Revenue, Cull Cattle Revenue. Outflows: Feed Cost, Water Cost, Energy Cost, Family Labor Cost (opportunity cost), Hired Labor Cost, Vet & Medicine, Other Operating Costs.

Auxiliary Variables (Converters): Milk Price (\$/cwt), Milk Yield (lbs/cow/yr), Water Use Rate (gal/cow/day), Energy Use (kWh/cow/yr), Electricity Rate (\$/kWh), Off-Farm Wage Rate (\$/hr for imputed family labor), Farm Size Class (small/mid/large switch).

Sources

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